Empirical distribution of site-determining ion matches under competing phospho-localizations

Testing AScore's uniform-fragmentation assumption on Prosit-Pioneer output

Pioneer.jl / phospho-localization work

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Motivation

The AScore method (Beausoleil *et al.*, *Nat. Biotechnol.* 2006, Fig. 3) scores phospho-localization certainty by computing, for each candidate site, a cumulative binomial probability of matching at least n of the top-i-per-100-m/z peaks across a peptide's N predicted b/y ions. The AScore is then the $-10\log_{10}$ difference in cumulative binomial probability, restricted to *site-determining ions* (those whose m/z differs between the top two candidate localizations), evaluated at the peak depth i that maximizes the top-two separation.

The step in AScore that we want to check is its **null model**: each predicted site-determining ion has match probability $p = i/100$ under H_0. This assumes

1. every predicted ion is *a priori* equally likely to be observed (uniform-fragmentation prior), and
2. the null-match probability depends only on the global top-i-per-100-m/z peak density, not on Prosit's intensity prediction or on local spectrum density.

In a Prosit-era pipeline, both assumptions are questionable: we can weight ions by predicted intensity (or restrict to the intensity-selected set already in the library), and we have a spectrum-level peak density we can measure directly. This note tests, empirically:

- Under a real Prosit-Pioneer search, how often does a site-determining ion *actually* match an observed peak, split by
  - whether the ion belongs to the true target's predicted fragment set or to a competing (loser) localization;
  - the number of distinguishing ions available for the pair;
  - the number of peaks in the pertinent MS/MS scan;
  - the number of possible localizations C for the base peptide.

Terminology

Throughout, a *pair* is one (target PSM, competitor localization) tuple.

- **Peptidoform.** A peptide sequence with a specific choice of PTM positions (e.g. `QSSVTQVTEQpSPK` — phospho on position 11).

- **Target.** The peptidoform that Pioneer's search actually reported as a passing PSM at some scan (best-per-precursor row in `precursors_long.arrow`, $q_{\text{value}} \leq 0.01$).
- **Competitor.** Any *other* peptidoform sharing the same base sequence and charge as the target but with phospho on different residues — i.e. a target-only positional isomer. Competitors are enumerated from a **clean library** (`yeast_plus_standards_1mc.poin`) with no shadow-site decoys, so a competitor is always another *target* peptidoform, never a designed decoy.
- **Distinguishing ion.** A b- or y-type fragment ion (charges 1 or 2) whose *predicted m/z* differs between the target and the competitor. Equivalently, the cumulative phospho count in the b/y ladder differs at that position. For b_i: target has $\sum_{k \leq i} \mathbb{1}[k \in \text{phos}]$ different from the competitor's; same for y_i counting from the C-terminus.
- **Site-determining ion.** AScore terminology, same as "distinguishing ion" here.
- **"Match".** An ion with predicted *m/z* m **matches** the observed spectrum if the spectrum has any peak within ± 20 ppm of m. Presence/absence only; intensity is not thresholded in this pass.

Two definitions of "available distinguishing ions" per pair are recorded:

- **`n_avail_theo`** — the count of *all* b/y ions at charges 1,2 whose cumulative phospho count differs between target and competitor. Bounded above by $2 \cdot 2 \cdot (L-1)$ for a peptide of length L.
- **`n_avail_lib`** — the subset of `n_avail_theo` that survives the Prosit intensity-based retention in the *target's* library-retained fragment list (`detailed_fragments.jls`). That list is capped near ~10 fragments per precursor and preserves site-determining coverage across mod-gap intervals, so `n_avail_lib` is small (typically 2–10) but each entry is an ion Prosit predicted with meaningful intensity.

The two "match rate" numbers, precisely

For each pair (t, c):

- n_{avail} = number of distinguishing ions considered (theoretical or library — recorded both ways).
- $n_{\text{match,tgt}}$ = number of those distinguishing ions where the *target's* predicted *m/z* has an observed peak within tolerance.
- $n_{\text{match,cmp}}$ = number of those distinguishing ions where the *competitor's* predicted *m/z* has an observed peak within tolerance.

The two counts share the same n_{avail} denominator (same set of ion indices $(b/y, i, z)$), but they are evaluated at *different m/z values*: the target's m/z uses the target's phospho positions, and the competitor's uses the competitor's.

Important — the competitor rate is NOT a clean null. In DIA the isolation window co-fragments any co-eluting precursors that fall within it, and positional isomers of the same base peptide have identical mass and near-identical LC retention time, so co-elution is the *norm*, not the exception. The competitor's shifted b/y lines can therefore genuinely be present in the same scan — not because the competitor localization is the truth for the target's PSM, but because a real co-eluting isomer contributed those peaks. The "competitor rate" reported below is therefore a **mixture**:

- (a) coincidental matches from spectrum density (what AScore's uniform-p null is trying to model), plus
- (b) real matches from co-eluting positional isomers that share the same precursor mass and RT.

Because component (b) is nonzero and cannot be separated from (a) without external information (e.g. whether the competitor localization was independently identified at nearby scans), the reported competitor rate is an **upper bound** on the true noise floor p_0 that a probability model would plug in as the wrong-localization null.

Two reported summary statistics, aggregated over all pairs in a stratum:

- **Target rate** (signal side): $\left\langle \frac{n_{\text{match,tgt}}}{n_{\text{avail}}} \right\rangle$. Under the hypothesis that the target's assigned localization is *at least one of the* true localizations present at that scan, this is the rate at which site-determining ions Prosit predicts for the target are actually observed.

- **Competitor rate** (mixture — see above): $\left\langle \frac{n_{\text{match,cmp}}}{n_{\text{avail}}} \right\rangle$. The rate at which the *shifted* m/z lines predicted by an alternative localization are observed. If the competitor is not co-eluting, this is the density-driven noise floor; if it *is* co-eluting, this includes real signal from the co-eluting isomer.

  Concretely: for target `QSSVTQVTEQpSPK` (phospho @ pos 11) vs competitor `QSSVTQVTEQSPpK` (phospho @ pos 13), the b_{11} ion is distinguishing — target predicts b_{11} at m/z x (heavier, includes phospho); competitor predicts b_{11} at m/z $x - m_{\text{phos}}/z$. If both peptidoforms co-elute and land in the same isolation window, *both* b_{11} lines will appear — the "competitor rate" then reflects real fragmentation, not chance density.

Data and method

Search

- MS data: 12 files from the Coon phospho dilution series (`dilution_curve2/combined`).
- Search: `dilution_curve2/combined/results` on branch `feature/localization-flr-scoring`.
- Search-time library: `yps_decoy2.poin` (yeast + Coon standards, 1 missed cleavage, with positional-isomer shadow decoys). The shadow decoys were only used at search time; the analysis restricts to target-vs-target pairs, so shadow-decoy competition does not contaminate the rates reported here.

Analysis library (clean, no positional-isomer decoys)

- `yeast_plus_standards_1mc.poin` — same peptide set as `yps_decoy2` but built with `is_loc_decoy = 0` throughout. 2,653,366 targets, of which 2,418,808 have $C > 1$.
- Precursor-space join between search library and analysis library is by (sequence, sorted phospho positions, sort so `precursor_idx` values are re-resolved.

Pair enumeration

1. Take the target-passing PSMs from `precursors_long.arrow` ($q_{\text{value}} \leq 0.01$, $q^{\text{global}} \leq 0.01$).
2. Re-key each PSM into the analysis library. Drop any that don't map.
3. For each remaining target PSM t, enumerate all competitors c from 'analysis_lib.grpT[(sequence_t, charge_t)] {t}$ — i.e. every target sibling positional isomer at that charge.
4. Sample 10,000 target PSMs uniformly at random (seed 1). This produced **156,715 pair rows** (mean 15.7 competitors per PSM).

Per-pair computation

For each pair (t, c) and each $(T, i, z) \in \{b, y\} \times \{1, \ldots, L-1\} \times \{1, 2\}$:

- Compute the cumulative phospho count on the target's ladder (cT) and the competitor's ladder (cC). If cT = cC the ion is *non-distinguishing* — skip.
- Otherwise compute the target's predicted *m/z*: sum of residue monoisotopic masses on the ion side of the cleavage + $c_T \cdot m_{\text{phos}}$
  - CAM $\cdot\, n_{\text{cam}}$, converted with the appropriate proton and (for y-ions) water contributions. Compute the competitor's predicted *m/z* the same way but with c_C.
- Match each *m/z* against the sorted observed peak list of scan `scan_idx` at ±20 ppm. Presence/absence only.
- Increment `n_avail_theo`; increment `n_match_tgt_t` and `n_match_cmp_t` per side if the corresponding m/z matched.
- If (T, i, z) appears in the target's library-retained fragment set (from `detailed_fragments.jls`), also increment `n_avail_lib`, `n_match_tgt_l`, `n_match_cmp_l`.

Also recorded per pair: scan peak count (`n_peaks_scan = length(mz_array[scan_idx])`), isolation-window peak count (peaks inside the scan's isolation window from `centerMz ± isolationWidthMz / 2`), C (number of possible localizations, $\binom{n_{\text{STY}}}{n_{\text{phos}}}$), and `iso_group_size_at_scan` (search-time competitor count including any shadow decoys).

Script

`dev_docs/prosit_ptm_integration/prototype/explore_matched_counts.jl`.

Output

Per-pair Arrow table at `/Users/n.t.wamsley/prosit_phospho_test/dilution_curve2/combined/results/exp` (156,715 rows).

Results

Numbers below are means over the pairs in each stratum.

Match rate vs number of distinguishing ions

Theoretical (`n_avail_theo`):

n_avail (bin)	n pairs	target rate	competitor rate
4	5,782	0.279	0.230
7–8	6,196	0.295	0.234
11–12	5,718	0.316	0.224
13–16	6,606	0.346	0.198
17–20	9,514	0.361	0.169
21–24	13,113	0.360	0.156
25–32	31,172	0.340	0.146
33+	78,614	0.297	0.139

Prosit library-retained subset (`n_avail_lib`):

n_avail (bin)	n pairs	target rate	competitor rate
2	14,780	0.691	0.290
3	16,602	0.710	0.277
4	17,739	0.736	0.260
5	17,362	0.757	0.264
6	18,204	0.774	0.257
7–8	31,771	0.780	0.253
9–10	19,079	0.784	0.252

The library-retained view lifts the target rate from ~30% to ~75% and drops n_{avail} from a median in the mid-20s to 4–8. The competitor rate is roughly flat around 26%.

Match rate stratified by scan peak count

scan peak count	n pairs	target rate	competitor rate
0 – 500	122	0.167	**0.029**
501 – 1000	2,355	0.192	0.056
1001 – 1500	12,188	0.236	0.098
1501 – 2000	30,310	0.280	0.127
2001 – 3000	83,579	0.322	0.162
3001+	28,161	0.387	**0.204**

Competitor rate spans ~7× across scans — from ~3% in sparse spectra to ~20% in the densest. A single global p cannot represent both regimes.

Match rate stratified by isolation-window peak count

iso-window peaks	n pairs	target rate	competitor rate
0 – 20	155,697	0.316	0.156
21 – 50	1,018	0.334	0.146

Effectively bimodal at ≤ 20 vs > 20 in this dataset, but the vast majority of pairs are in the low bin — the isolation-window peak count is a weaker covariate than the whole-scan count.

Match rate stratified by C (number of possible localizations)

C	n pairs	target rate	competitor rate
2	2,602	0.365	0.175
3	9,098	0.356	0.172
4 – 6	58,592	0.347	0.164
7 – 10	57,165	0.311	0.148
11 – 20	16,138	0.272	0.141
21+	13,104	0.224	0.161

Target–null gap narrows monotonically with C: 2.1× at $C = 2$ down to 1.4× at $C \geq 21$.

Fraction distribution

Histogram of $n_{\text{match,tgt}}/n^{\text{theo}}_{\text{avail}}$ across all 156,715 pairs:

bin	n pairs	%
[0.0, 0.1)	9,007	5.7
[0.1, 0.2)	27,380	17.5
[0.2, 0.3)	44,465	28.4
[0.3, 0.4)	33,932	21.7
[0.4, 0.5)	19,248	12.3
[0.5, 0.6)	16,593	10.6
[0.6, 0.7)	5,760	3.7
[0.7, 0.8)	2,101	1.3
[0.8, 0.9)	376	0.2
[0.9, 1.0)	141	0.1

Right-skewed, mode at 0.2–0.3, tail out to 1.0. Not a single-p binomial.

Interpretation

1. **The observed competitor rate is not a clean null, but even its upper bound is inconsistent with AScore's uniform-p assumption.** The empirical competitor rate is 14–25% on the theoretical distinguishing set and 25–29% on the Prosit-retained subset — orders of magnitude above the $i/100$ figure AScore's binomial applies (~1–10%). Because that rate mixes coincidental matches with real co-eluting-isomer signal (see §"The two 'match rate' numbers, precisely"), the *true* noise-floor p_0 is lower, but a probability model still needs a data-driven p estimated from data where co-elution is either absent or explicitly modeled.
2. **`n_avail_lib` is the right definition for a probability model.** It gives:
   - Higher target-side match rate (~0.75 vs ~0.30) — each retained trial is more informative,

- Comparable competitor-side rate (0.26 vs 0.20 depending on stratum) — the null doesn't get much worse,
- Sharper signal:null contrast (~3:1 vs ~2:1),
- Fewer trials per pair (median 4–8 vs 25+) — a simpler distribution to calibrate.

3. **Peak-count covariate is real and large.** Competitor rate ranges from 3% in sparse spectra to 20% in dense ones. Some of that spread is density-driven noise (component (a) above); some is that dense spectra are more likely to contain co-eluting isomers whose real fragments push the mixture up. Either way, any probability we compute needs a peak-count covariate; a single global p under-reports ambiguity in dense scans and over-reports in sparse ones.

4. ***C* matters, but as difficulty, not a fundamentally different null.** The target-null gap narrows with C because high-C peptides have more distinguishing ions that collide with real fragment lines from other positions, not because the null pattern changes. A probability model can handle this by simply using C as a covariate on the same ion-level null.

5. **The empirical fraction distribution is not binomial-uniform.** It is right-skewed with a heavy tail at high fractions — consistent with a mixture of true localizations (high fraction) and wrong or ambiguous localizations (low fraction). A likelihood-ratio model that uses per-ion Prosit intensity predictions is a more natural fit than a single-p binomial.

Caveats

- **Competitor rate is a mixture, not a noise floor.** As noted in the "match rate" section, positional isomers of the same base peptide co-elute and are co-fragmented in the same DIA isolation window, so the shifted m/z lines that define the competitor rate can be real, not coincidental. The reported competitor rate is therefore an upper bound on the true wrong-localization noise floor p_0. The next-step analyses below address this directly.

- **Sample size.** 10,000 PSMs from the 969,006 target-passing PSMs across 12 files. Distributions are already very stable — bins with $\geq 5{,}000$ pairs converge to ± 0.005 on a stratum-mean rate — but rerunning at 50k would tighten low-count bins (e.g. `scan peak count 0-500`, $n = 122$).

- **"Match" is presence/absence at 20 ppm.** No intensity threshold. Adding an intensity threshold (e.g. observed peak intensity $\geq$ 1% of base peak) would lower both target and competitor rates; the ratio is what matters and is unlikely to change much.

- **`n_avail_lib` uses the target's library-retained set, not the competitor's.** Symmetry could be enforced (union or intersection of the two retained sets) but the target's set is what a real localization scorer would use (we're testing whether target = true; competitor's Prosit prediction is not obviously the right basis).

- **Competitor sampling.** For high-C peptides, every sibling positional isomer is treated as a competitor and each contributes a pair — so the pair count is dominated by pairs from a few high-C peptides. Per-pair stratification is honest; per-PSM stratification (mean across pairs first, then across PSMs) would rebalance and is a straightforward re-summary of the same Arrow.

- **Search-time competition was against `yps_decoy2`, not the clean library.** This affects which PSMs Pioneer chose as best-per-precursor (some target PSMs may have lost weight to shadow decoys), but does not affect the spectrum-space match rates recomputed here.

Files

- **Script.** `dev_docs/prosit_ptm_integration/prototype/explore_matched_counts.jl`
- **Per-pair Arrow.** `/Users/n.t.wamsley/prosit_phospho_test/dilution_curve2/combined/results/ex` (156,715 rows).
- **Related prior work.**
  - `dev_docs/prosit_ptm_integration/prototype/extract_sds_features.jl` — the original site-determining intensity-ratio feature extractor.
  - `dev_docs/prosit_ptm_integration/prototype/PROTOTYPE_STATUS.md` — the S-score validation status prior to this exploration.
  - AScore paper: Beausoleil *et al.*, *Nat. Biotechnol.* 24(10), 1285 (2006).

Next step candidates

1. **Split the competitor rate into co-eluting vs not-co-eluting.** Two approaches on the same Arrow, giving a cleaner estimate of the true noise floor p_0:

   a. **Presence filter.** For each pair (t, c), mark the competitor as "co-eluting" if the competitor precursor c was itself reported as a passing PSM (any q-value bin) at any scan within $\pm\Delta$ RT of t's scan in the same file. Recompute the competitor rate separately on co-eluting and non-co-eluting subsets. The non-co-eluting subset approximates the AScore-analog uniform-p null.

   b. **Weight filter.** If per-scan per-precursor deconvolution weights can be dumped (existing `DIAG_DUMP_FILE_IDX` hook — see `SearchDIA/SearchMethods/MainSearch/scoring.jl:89-90`), condition on the competitor's `weight` at the *target's* scan being essentially zero. This is a sharper filter than presence-at-nearby-scan because it uses the search's own co-elution estimate.

2. **Draft a per-ion p_i model on the co-elution-corrected null.** Once p_0 is cleanly estimated, model per-ion match probability as $p_i = f(\text{local peak density}, \text{predicted intensity})$; restricted to `n_avail_lib`. Weighted Poisson-binomial tail as the analog of AScore's cumulative binomial, evaluated per pair, and produce a per-target localization score $= -10 \log_{10}(P_c/P_t)$.

3. **Rebalance to per-PSM.** Same Arrow, different summary: mean across competitors per PSM, then mean across PSMs, stratified same as above. High-C peptides currently dominate the pair count.

4. **Add an intensity threshold on "match".** Vary `> 0.01 * base peak` or `> 0` and confirm the ratio is stable. Report competitor rate at rank thresholds equivalent to AScore's $i \in \{1, \ldots, 10\}$ to draw a direct comparison against AScore's uniform-p assumption.